

Find the Fourier transform of the following functions $f: \mathbb{R} \rightarrow \mathbb{R}$

a)

$$f(x) = e^{-2|x|}$$

$$\begin{aligned}\hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-2|x|} e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 e^{2x} e^{-i\omega x} dx + \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-2x} e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 e^{x(2-i\omega)} dx + \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-x(2+i\omega)} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{x(2-i\omega)}}{2-i\omega} \right]_{-\infty}^0 + \frac{1}{\sqrt{2\pi}} \left[-\frac{e^{-x(2+i\omega)}}{2+i\omega} \right]_0^{\infty} \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{e^{0(2-i\omega)}}{2-i\omega} - \frac{e^{-\infty(2-i\omega)}}{2-i\omega} \right) + \frac{1}{\sqrt{2\pi}} \left(-\frac{e^{-\infty(2+i\omega)}}{2+i\omega} - \left(-\frac{e^{-0(2+i\omega)}}{2+i\omega} \right) \right) \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{2-i\omega} + \frac{1}{2+i\omega} \right) \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{2+i\omega}{2-i\omega} + \frac{2-i\omega}{2+i\omega} \right) \\ &= \frac{1}{\sqrt{2\pi}} \frac{4}{4+\omega^2} \\ &= \sqrt{\frac{2}{\pi}} \frac{4}{4+\omega^2}\end{aligned}$$

b)

$$f(x) = \begin{cases} e^x, & x \in [-a, a] \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned}\hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a f(x) e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a e^x e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a e^{x(1-i\omega)} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{x(1-i\omega)}}{1-i\omega} \right]_{-a}^a \\ &= \frac{1}{\sqrt{2\pi}} \frac{e^{a(1-i\omega)} - e^{-a(1-i\omega)}}{1-i\omega}\end{aligned}$$

c)

$$f(x) = x^2 e^{-x^2}$$

$$g(x) = e^{-x^2} \Rightarrow g'(x) = -2x e^{-x^2} \Rightarrow g''(x) = 4x^2 e^{-x^2} - 2e^{-x^2}$$

Then we can rewrite the expression for $f(x)$ using $g(x)$ and $g''(x)$

$$f(x) = \frac{g''(x) + 2g(x)}{4} \Rightarrow F(f) = \frac{F(g''(x) + 2F(g(x)))}{4} \Rightarrow F(f) = \frac{-\omega^2 F(g(x)) + 2F(g(x))}{4}$$

We know that

$$F(f'(x)) = i\omega F(f(x))$$

Hence

$$F(f''(x)) = -\omega^2 F(f(x))$$

$$\begin{aligned} F(e^{-ax^2}) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ax^2} e^{i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ax^2 + i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a(x^2 - \frac{i\omega x}{a})} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a(x^2 - 2x \cdot \frac{i\omega}{2a})} dx \quad x^2 - 2ax = (x-a)^2 - a^2, \quad a = \frac{i\omega}{2a} \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a(x - \frac{i\omega}{2a})^2 - (\frac{i\omega}{2a})^2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a(x - \frac{i\omega}{2a})^2} e^{\frac{\omega^2}{4a^2}} dx \\ &= \frac{1}{\sqrt{2\pi}} e^{\frac{\omega^2}{4a^2}} \int_{-\infty}^{\infty} e^{-a(x - \frac{i\omega}{2a})^2} dx \quad u = \sqrt{a}(x - \frac{i\omega}{2a}) \Rightarrow \frac{du}{dx} = \sqrt{a} \Rightarrow dx = \frac{du}{\sqrt{a}} \\ &= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{a}} e^{\frac{\omega^2}{4a^2}} \int_{-\infty}^{\infty} e^{-u^2} du, \quad \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \\ &= \frac{1}{\sqrt{2\pi}} \cdot \frac{\sqrt{\pi}}{\sqrt{a}} e^{\frac{\omega^2}{4a^2}} \\ &= \frac{1}{\sqrt{2a}} e^{\frac{\omega^2}{4a^2}} \end{aligned}$$

$$F(x^2 e^{-x^2}) \Rightarrow F(f) = \frac{-\omega^2 F(g(x)) + 2F(g(x))}{4} = \frac{-\omega^2 F(e^{-x^2}) + 2F(e^{-x^2})}{4} = \frac{-\frac{\omega^2}{4\sqrt{2}} e^{-\frac{\omega^2}{4}} + \frac{2}{2\sqrt{2}} e^{-\frac{\omega^2}{4}}}{4} = \frac{e^{-\frac{\omega^2}{4}} (\frac{1}{2\sqrt{2}} - \frac{\omega^2}{4\sqrt{2}})}{4}$$

Prob. 2

Define the functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ as

$$f(x) := e^{-x^2} \quad \text{and} \quad g(x) := e^{-2x^2}$$

Use the Fourier transform in order to compute the convolution $f * g$

$$f(x) = e^{-ax^2} \xrightarrow{\text{Proved this in 1c)}} \hat{f}(\omega) = \frac{1}{\sqrt{2a}} e^{-\frac{\omega^2}{4a}}$$

$$F(f) = \frac{1}{\sqrt{a}} e^{-\frac{\omega^2}{4}}$$

$$F(g) = \frac{1}{2} e^{-\frac{\omega^2}{8}}$$

$$F(f * g) = \sqrt{2\pi} F(f) \cdot F(g)$$

$$= \sqrt{2\pi} \frac{1}{\sqrt{2}} e^{-\frac{\omega^2}{4}} \frac{1}{2} e^{-\frac{\omega^2}{4}}$$

$$= \frac{\sqrt{2}}{2} e^{-\frac{3\omega^2}{8}}$$

$$(f * g)(x) = F^{-1}\left(\frac{\sqrt{2}}{2} e^{-\frac{3\omega^2}{8}}\right)$$

$$= \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x} d\omega$$

$$= \frac{\sqrt{2}}{2} \int_{-\infty}^{\infty} e^{-\frac{3\omega^2}{8}} e^{i\omega x} d\omega$$

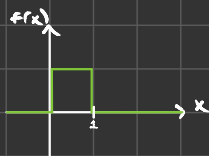
$$= \frac{\sqrt{2}}{2} \cdot \frac{1}{\sqrt{2 \cdot \frac{3}{8}}} \cdot e^{-\frac{\omega^2}{2 \cdot \frac{3}{8}}}$$

$$= \sqrt{\frac{2}{3}} e^{-\frac{\omega^2}{6}}$$

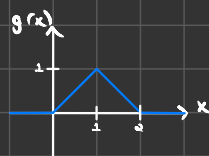
Prob. 3

Def. the func. $f, g: \mathbb{R} \rightarrow \mathbb{R}$ as

$$f(x) := \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{else} \end{cases}$$



$$g(x) := \begin{cases} x & \text{if } 0 < x \leq 1 \\ 2-x & \text{if } 1 < x < 2 \\ 0 & \text{else} \end{cases}$$



a) Use the def of convolution in order to show that $f * f = g$

$$f * f = \int_0^1 f(x-\tau) f(\tau) d\tau = \int_0^1 1 d\tau = [\tau]_0^1 = 1$$

$$g(x) = \int_0^1 x dx + \int_1^2 2-x dx = \left[\frac{1}{2} x^2 \right]_0^1 + \left[2x - \frac{1}{2} x^2 \right]_1^2 = \frac{1}{2} + (4-2 - (2 - \frac{1}{2})) = 1$$

We can see that $f * f = 1$ and that g in its domain corresponds to 1.

b) Compute the Fourier transform of f .

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^1 e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[-\frac{e^{-i\omega x}}{i\omega} \right]_0^1$$

$$= \frac{1}{\sqrt{2\pi}(\omega)} (-e^{-i\omega} + 1)$$

c) Compute the Fourier transform of g .

$$\begin{aligned}\hat{g}(\omega) &= F(g) \\ &= F(f * f) \\ &= \sqrt{2\pi} F(f) F(f) \\ &= \frac{1}{i\omega} (-e^{i\omega} + 1)^2\end{aligned}$$

Prob 4

a) Let $\omega_N := e^{\frac{2\pi i}{N}}$ for some $N \in \mathbb{N}$. Show that

$$\sum_{k=0}^{N-1} \omega_N^{jk} = \begin{cases} N, & \text{if } j \bmod N = 0 \\ 0, & \text{otherwise} \end{cases}$$

b) Show that

$$\sum_{k=0}^{N-1} e^{\frac{2\pi i n k}{N}} e^{\frac{2\pi i m k}{N}} = \begin{cases} N, & n=m \\ 0, & \text{otherwise,} \end{cases}$$

for some $n, m \in \{0, \dots, N-1\}$

c) Compute explicitly the Fourier matrices F_2 , F_3 and F_4 .

$$F_{jk} = e^{\frac{2\pi i j k}{n}}, \quad j, k = 0, 1, 2, \dots, n-1$$

$$F_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} F_{0,0} & F_{0,1} \\ F_{1,0} & F_{1,1} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} e^{\frac{2\pi i 0 \cdot 0}{2}} & e^{\frac{2\pi i 0 \cdot 1}{2}} \\ e^{\frac{2\pi i 1 \cdot 0}{2}} & e^{\frac{2\pi i 1 \cdot 1}{2}} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & e^i \end{bmatrix} = \underline{\underline{\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}}$$

$$F_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} F_{0,0} & F_{0,1} & F_{0,2} \\ F_{1,0} & F_{1,1} & F_{1,2} \\ F_{2,0} & F_{2,1} & F_{2,2} \end{bmatrix} = \underline{\underline{\frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & e^{\frac{2\pi i}{3}} & e^{\frac{4\pi i}{3}} \\ 1 & e^{\frac{4\pi i}{3}} & e^{\frac{8\pi i}{3}} \end{bmatrix}}}$$

$$F_4 = \frac{1}{\sqrt{4}} \begin{bmatrix} F_{0,0} & F_{0,1} & F_{0,2} & F_{0,3} \\ F_{1,0} & F_{1,1} & F_{1,2} & F_{1,3} \\ F_{2,0} & F_{2,1} & F_{2,2} & F_{2,3} \\ F_{3,0} & F_{3,1} & F_{3,2} & F_{3,3} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & e^{\frac{2\pi i}{4}} & e^{\frac{2\pi i \cdot 1 \cdot 2}{4}} & e^{\frac{2\pi i \cdot 1 \cdot 3}{4}} \\ 1 & e^{\frac{2\pi i \cdot 2 \cdot 1}{4}} & e^{\frac{2\pi i \cdot 2 \cdot 2}{4}} & e^{\frac{2\pi i \cdot 2 \cdot 3}{4}} \\ 1 & e^{\frac{2\pi i \cdot 3 \cdot 1}{4}} & e^{\frac{2\pi i \cdot 3 \cdot 2}{4}} & e^{\frac{2\pi i \cdot 3 \cdot 3}{4}} \end{bmatrix} = \underline{\underline{\frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}}}$$

d) Compute by hand the discrete Fourier transform of the vector $f = [1, 2, 3, 3]$

$$F_4(f) = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 2 & 3 & 3 \\ 1 & -2i & -3 & 3i \\ 1 & -2 & 3 & -3 \\ 1 & 2i & -3 & -3i \end{bmatrix} = \underline{\underline{\begin{bmatrix} 10 \\ -1 + 2i \\ -2 \\ -1 - 2i \end{bmatrix}}}$$